

Tutorial 3 Answer Key

1. 2.67 A

2. 9.6 V

3. 10 Ω

4. a. Current in $C_1 = 0$, Current in $R_1 = 0$, Current across $C_1 = V$

b. $R \frac{di}{dt} + i \left(\frac{1}{c_1} + \frac{1}{c_2} \right) = 0$

c. $i(t) = \frac{v}{R} e^{-\frac{2t}{RC}}$

5. $i(t) = \frac{V}{R_1} e^{-(R_1+R_2)t/L}$

6. a. 6 μs b. 250 ms

7. $i(t) = -6e^{-20t}$

8. a. 28.84 H b. 33.64 H

9. The wheel with larger time constant will be easier to pedal.

10. $V_c(t) = 8e^{-2t}$

11. $i(t) = 2e^{-2t}$